Patents

Hydrogen gas injector for internal combustion engine

Abstract

System and apparatus for the controlled intermixing of a volatile hydrogen gas with oxygen and other non-combustible gasses in a combustion system. In a preferred arrangement the source of volatile gas is a hydrogen source, and the non-combustible gasses are the exhaust gasses of the combustion system in a closed loop arrangement. Specific structure for the controlled mixing of the gasses, the fuel flow control, and safety are disclosed.

Classifications

F02M25/12 Engine-pertinent apparatus for adding non-fuel substances or small quantities of secondary fuel to combustion-air, main fuel or fuel-air mixture adding acetylene, non-waterborne hydrogen, non-airborne oxygen, or ozone the apparatus having means for generating such gases

View 2 more classifications

Landscapes

Engineering & Computer Science

Chemical & Material Sciences

Show more

JPS58202352A

Japan

Find Prior Art Similar

Other languages: Japanese

Inventor: スタンリー・エイ・メイヤー

Current Assignee : Individual

Worldwide applications

1982 ~~US~~ 1983 ~~CA~~ ~~DE~~ ~~EP~~ ~~AT~~ JP

Application JP58023667A events

1983-02-15 Application filed by Individual

1983-11-25 Publication of JPS58202352A

Status Pending

Info: Patent citations (14), Cited by (33), Similar documents, Priority and Related Applications

External links: Espacenet, Global Dossier, Discuss

Claims (1)

Hide Dependent

translated from Japanese

[Claims] (!) A housing with a reservoir for retaining natural water and gas collection vines for maintaining a predetermined pressurized gas: a pair of similar reservoirs disposed within said reservoir; Non-oxidizing plate: and a DC rolling mill connected to the above plate to dissociate hydrogen and oxygen atoms from the water molecules.

[a hydrogen/oxygen generator comprising a flow source; a gas mixing chamber; means for connecting a water-based gas obtained from said hydrogen generator to said mixing chamber; and regulating the flow of hydrogen gas to said mixing means. a linear fuel control valve disposed in said hydrogen gas connection means for said port and including said boat and an annular element disposed within said port and connecting said opening thereof; a non-volatile gas supply source; means for connecting a non-volatile gas obtained from a non-volatile gas source to said mixing chamber; and means connected to an inlet of said mixing chamber for merging air with said hydrogen gas and said non-volatile gas. Confinement/111 Air intake means and a gas burner using a predetermined amount of hydrogen gas supplied from the above-mentioned non-volatile gas and the air from the above-mentioned air intake means. Combustion equipment on standby. (21) The field air intake means has a housing, and a variable position plate valve is disposed on the housing.

(31) The combustion apparatus according to claim 11, wherein the air intake lid to the mixing pipe of e is controlled. 14+ The combustion device according to claim 11, further comprising a housing having a middle four-port, and said cross-shaped element is located in said port. The combustion device according to claim (1), which is a structure having a chi/4. Claim 1 comprising means for displacing the first and second structures of the first and second parts.

The combustion device according to item 41. +6+ The combustion apparatus according to claim 10, wherein the mixing chamber has a cooling device at its hydrogen gas inlet and a quenching device at its nonvolatile gas inlet. (7) The above-mentioned gas mixing chamber has a trap area in its upper glum region, and the hydrogen gas connection to the upper one joint chamber is located in the uppermost area of the above-mentioned trap area.

The combustion device according to item 11. (81) The above-mentioned ambient air intake section is the combustion device of Claim (7) Ring 8r2, which is located in the lowest boundary area of the above-mentioned truck. (9) The above-mentioned ambient air intake means is provided with a control valve. , the non-flammable gas intake means is provided with control valves, these valves are freely operable to introduce into the mixing chamber a mixture of air and non-flammable gas in a fixed ratio of Pis. 1111 The above-mentioned air intake means releases excess hydrogen in the trap area of -Fae in the combustion chamber and reduces the amount of hydrogen in the combustion chamber. The combustion device according to item (8). A control valve is provided to fit the parbu of OuF and the hydrogen gas M! connection means of upper 6C, and the on/off fm of the combustion device is provided.

d. The combustion device according to claim 1, wherein the gas flow is controlled during the period d. The control valve connected to the burner of n's tomr4 is

The combustion arrangement according to claim 71, which is a pressure regulating chamber. Housing means surrounding a mixing chamber of 0-hysteresis and a first opening provided in the housing for air injection. and an aperture provided in the σ housing for releasing hydrogen gas trapped in the σ housing. The upper enclosed area of your own hydrogen gas generator is

A pressure release valve f that operates when hydrogen gas held therein exceeds a predetermined voltage:

The combustion device described in item 1) is connected to the power source and the pressure release valve, and when the combustion device is inoperable, the hydrogen gas generator is connected to the voltage source and the pressure release valve. The combustion device according to item A4 of the +4f permission requirement, which is equipped with a sensing switch that operates to shut off the wire. aS The means for connecting the hydrogen gas to the mixing chamber is provided with a one-way check valve, Claim I11 g1 2-mount combustion device. [I? 1. The combustion apparatus according to claim 11, wherein the means for connecting the hydrogen gas to the mixing chamber includes a quenching device. 112. The combustion of claim 111, wherein as said gas burner is an internal combustion engine and said source of non-volatile gas is the exhaust gas of said engine. Device. Combustion 11 according to claim 10, comprising: an oil supply source and an oil spray connected to the oil supply source and the mixing chamber.

Description

translated from Japanese

[Detailed description of the invention] This book relates to a combustion device.

The present applicant filed an application for "Hydrogen Generation/Stem" Q on September 76, 2007. Beiyuan Patent Application No. 302,
No. g07 discloses a hydrogen generation system that converts water into hydrogen and oxygen gas.

In the system disclosed in this mountain axis, water is passed between four non-oxidizing gold I/I plates of one kind, and these metal plates are subjected to a DC voltage/current of non-tuning, non-tonami, and one-power. By applying , hydrogen atoms are dissociated from water molecules. By converting this non-V@regular, non-Tonami DC masticating pressure /@(& into a Noculus, sub-atomic (sub
-atomic) action is promoted.

The apparatus includes various configurations for separating the generated hydrogen from oxygen gas.

In addition, Motoyama filed an application on September 5, 1997 for [Hydrogen Air Desimin Processing Equipment] in the U.S. Special Rods No. 6, Kuguda, which uses non-elastic, royal gas. An apparatus is disclosed in which a volatile gas is controlled with a volatile gas in the mixing stage. This hydrogen aeration treatment system (h ydragsnalrdatlon processo
r system) utilizes a rotary, waxy gas displacement device to charge, meter, mix, and pressurize the gas. When injecting the gas, ambient air is passed through the open frame gas combustion device to remove any gas or other substances present. Thereafter, the non-flammable gas mixture is cooled and mechanically mixed with F and a certain amount of hydrogen gas to remove impurities. In this way, new synthesis gas is produced. The rotary mechanical gas displacement device described above measures the volume of synthesis gas thus produced and determines the correct gas mixture ratio to produce the desired combustion rate of hydrogen gas. Determine the volume of synthesis gas that should be used.

-Hydrogen Air Duration Process According to U.S. Patent Application No.
The vjL is a multi-stage system useful in a variety of special applications. Also, the other tree [J+! +Patent Application Hydrogen Generation System discloses a unique hydrogen generation device that is only 5kjli.

In addition, the present applicant filed L U.S. Watch Application 2!37 on October 7g, /9g/. ;, No. 993 discloses a combustion device that can be applied to a mechanically pumped device. The combustion device W is useful, for example, for driving a screw I in a private vehicle. This application discloses a hydrogen generator that produces hydrogen and non-volatile gases (eg, 1!l! and J4$). A non-volatile gas is injected into the hydrogen gas tank, one flow path and an air intake provided with control means, so that the hydrogen, non-volatile gas and air are combined and mixed with each other. After that, the exhaust gas of the combustion chamber is sent to the combustion chamber and ignited there, and the exhaust gas of the combustion chamber is mixed with R4 gas as a non-flammable gas, thus forming a closed loop. The application discloses a specific implementation of such a system such as VX+4.

Accordingly, the present invention provides a combustion engine using hydrogen and non-flammable gas.
It is intended to provide a mixture of hydrogen and non-flammable gas (7,
An object of the present invention is to provide a combustion device which controls combustion temperature thereby.

Yet another object of the present invention is to provide a combustion system that reduces the flow of fuel into the combustion chamber by 1°, especially when using hydrogen gas.

Other objects and features of the invention will become apparent from the description below.

A particularly preferred embodiment of the device of the present invention is a furnace for driving a piston in a combustion device utilizing hydrogen gas, particularly an automatic W# device.

The apparatus of the present invention uses a hydrogen gas generator for generating hydrogen gas. Hydrogen gas and non-volatile gas are sent to a mixing chamber, and beak material is also sent to this mixing chamber. A mixture 6 of these gases is controlled to adjust the combustion temperature. That is, the combustion temperature and speed of hydrogen gas are controlled to be lowered by the combustion temperature of commercially available fuel a II (, i. kt).

The air intake becomes the oxygen source, and the air intake is also
, W has been done. Exhaust gas from the combustion chamber is controlled and utilized as a non-flammable gas. The nitrogen generator has a holding tank to provide a starting fuel source. The hydrogen gas generator also has a power switch that is activated from one position to the cabinet position by the action of a sensing switch above the combustion chamber.

The metered arrangement is provided with a bank of one-way valves, safety valves and cooling devices. A combination of these devices can be used to fuel a car engine (or any other fuel)
Construct a pre-assembled assembly for converting from hydrogen to hydrogen gas mixtures.

Hereinafter, the present invention will be explained along with the figure ([I].

FIG. 7 shows the entire gas mixing and fuel flow primary system utilized in a JP Akatsuki engine, particularly an engine utilized in a motor vehicle.

In FIG. 7, the hydrogen supply source is the hydrogen generator 10 disclosed and explained in the above-mentioned patent application by Tokusei Shipeng. The device accommodates 10 water tanks 2.

Water W! 2, s to the US patent application of the previous applicant
The first plates 3 are arranged in rows.

A direct current, pressure/l current, is applied to the coil 8 by opening the electric power section 27.

The upper part 7 of the device 10 is a water bulk storage area in which the required amount of hydrogen at pressure is maintained.

In this way, a temporary flow of water-based gas occurs at the beginning of wJ. To replenish the consumed water, the generator has a continuous water supply l. Thus, the occurrence #
ctIILFi is operated as described in the aforementioned patent application.

The safety valve 28 will fail if excess gas occurs.

On the other hand, the switch 26 is a gas pressure switch that operates the hydrogen generator when the gas volume decreases to maintain the gas pressure at a constant I5T level.

The generated hydrogen gas 4Fi is introduced into the gas mixing chamber 20 from the stop valve 16 to the gas mixing chamber 20, where the hydrogen gas is sent through the rPtL complex from the supply source described below. mixed with sexual gases.

In the unlikely event that the one-way valve 75 malfunctions, it will be difficult to ignite the bulk gas 4 in the storage area 7 of the hydrogen generator 11O.
In such a case, the spark generator 76 extinguishes the spark to prevent such ignition noise.

To explain this with reference to the second example, the bulk gas passes through the ηC path 5'kA, and the nonflammable gas passes through the flow path 9 to the carburetor, that is, the gas mixing chamber 2o. The gas mixing chamber further includes an ambient air intake 14.

The water-based gas 4 passes through the channel 5 and through the nozzle 11 into the trap area 46 of the mixing chamber 2o as sugley 16'. The nozzle 11 has an opening smaller than the opening in the plate in the cooling device 37, and has a suction hole smaller than the opening in the plate in the cooling device 37. - Prevent flashbacks from occurring in the event of a failure.

The non-volatile gas is injected through the nozzle 13 as soets gray 17 into the trap area 47 of the mixing chamber 2o. The emergency device 39 operates in the same manner as the rapid cooling device 37.
160 Aya Hmen A is 1 good 1. In the case of a strong inclusion mode, it becomes similar to the CLJT system, which is essential for the combustion of water-based gas. Furthermore, the hole opened in the above-mentioned shaft is a non-volatile gas, and the central part is the exhaust gas in the closed loop system. Of course - 1t
The IRX and/or non-volatile gases Q can also be supplied from independent sources.

Further, referring to FIG. 2, the gas drag area 47 has a size of 19T feet. Since the water system is fighting, the water system will rise and become trapped within area 47. The size of the area 47 is such that it can contain enough water bulk gas to ignite within a short period of time when the fuel engine is started.

It will be appreciated that the water-based gas is injected into the uptake area of the trap zone 47. The water rate is a volatile gas)
However, it ascends at a very fast speed (11m, 3x f or faster). Therefore, when hydrogen gas enters the mosquito low habituation area of the drag zone 47 (mixing zone), the hydrogen gas enters the air I! It cannot be mixed with the lk system and will rise rapidly.

In the drag section 1 shown in FIG. That is, the water-based gas t is forced downward into the air, and is easily mixed with it.

The non-flammable gas and ambient air (oxygen) passing through the flow paths 9 have a specific controlled ratio, determined for each engine. #4 95 cubic meters of non-flammable gas
Set the valve 45 to change the amount of air.
5II! Once the correct firing speed k is determined by adjusting the ratio, the ratio is maintained.

In systems where the non-flammable gas is the exhaust gas of an engine with a closed roof configuration, and the nitrogen gas is controlled by the ennon, the flow rate and therefore the mixture of chili/non-flammable gas is reduced by the acceleration of the engine. be done.

A mixture of air and incompatible gases serves as a carrier for hydrogen gas. That is, the water-based gas is combined with the nitrogen/non-flammable gas mixture. As the water-based gas is combined with the smoke/non-flammable gas mixture, the rotational speed of the engine is increased.

Next, the third evil will be explained. Figure 3 shows the details of fuel 9's axle load/type 1 flill m iron metal 53 in a new song.

The water bulk gas 4 passes through the gas population 41 and enters the chamber 43. Hydrogen gas leaves chamber 43 and travels upstream from port 42 to chamber 47'. The reservoir of gas entering chamber 47' from ♀43 is controlled by restricting the opening of port 42 (2).

Control is achieved by inserting a bit 73 with an Ayasugi taper into the opening of the hole. The blunt end of pin 73 is fixed to rod 71. The Rondo door 1 passes through the supporting cylinder 75' and is connected to the housing 30 by the first rotor 8.
1t- street, +#IL+--Shinsu 83 is carrying it.

The spring 49 connects the rod 71 with the pin 73 and opens the opening 4.
Hold against 2. When mechanism 83 is actuated, pin 73
is withdrawn through opening 42, increasing the gas 11 passing from 43 to chamber 47.

Stops 67, 69 hold spring 49 directly in position i>'. One rectangular position of the pin 73 relative to the opening 42 is adjusted by a nut 63,67 threaded onto the threaded rod 61. That is, by adjusting the screw, the idle speed is controlled and the minimum oscillation (ly gas is caused to flow from the shore 43 to the chamber 47) in order to continuously operate the combustion engine.

In addition to the porridge, a nitrogen-containing device is shown for manipulating the amount of air delivered to the mixing chamber 20.

One kzl layered on the plate 18 has h on its end 11'.
Mouth 7゛ [7]. A control &42' is slidably mounted above the shadow opening. - The grommet 12 (i-) is operated by the low electric current of the control rod 19 connected to the control board 11m to the control plate 1 relative to the mouth. In the event of an unaccompanied movement, the release valve 24 will break.

As shown in Part 9, if hydrogen gas builds up to excessive pressure in the mixing chamber 20, the port 34 on the hood 32 of the vehicle
A relief tube 36 connected to the X gas can safely escape into the atmosphere.

In the event that a single movement of combustion occurs within the mixing chamber 20, the pressure pipe all valves 33 are broken and hydrogen gas is expelled without combustion.

FIG. 3 shows a gas control system that can be replaced in an internal combustion engine of a mechanical motor vehicle without making any changes or modifications to the main meter or controller in the design of the motor vehicle.

Since the flow of hydrogen (extensible gas) is of course small, a gas flow f6 is provided in the flow path 5 to reduce the flow of hydrogen.
A t, l-valve 53 (FIG. 1) is provided. The quietness of the valve 53 is illustrated in FIG.

The total air intake section 14 can also be provided in a carburetor assembly having an intake section 55 that adjusts the opening of the plate 42', as explained in conjunction with FIG.

In order to maintain a constant pressure f in the water rod gas IFr zone during on-off operation of the engine, the gas related control valve is responsive to an electrical shutoff control switch 33'.

Controlling the foot pressure in this manner allows for an abundant supply of gas at start-up, as well as an even more abundant supply of gas during certain periods of operation.

Switch 33', on the other hand, is responsive to true full control switch 60. When the engine is running, the power is generated, and this real power creates a lead W 60a and passes through the negative switch 60, keeping the switch 33' in the state. When the engine is not running, the decreases and becomes low, and the switch 60
'i [then switch 33'kfif! IJ [let me,
The flow of hydrogen gas to the control valve 53 is stopped.

When a low peak a'dL pressure/'Kaku is applied to the safety 9f28, the solenoid 29 is activated. The solenoid connects the pressure switch 26 and the assembly terminal 27 to excite the hydrogen generator.
A control voltage is applied to the plate 3 serving as the device.

'When the solenoid 29 is actuated by electric power, hydrogen gas follows DE, rL, and the control valve 16, and becomes available for use.

The differential pressure output of hydrogen gas for I-series mixer 20 is FIJ
Yes, 'J-5ky or 1. -'75 4C, 3
0 pounds to 75 bonds). Water = X raw #C equipment 1
° is M2I! When the Il gas ratio level is reached, the pressure switch 2BR will cut off the power supply to the water system. If the pressure in the chamber exceeds the ar"JOE level, the safety release valve 28
is activated, disconnects the #IL flow, and shuts down the entire system for safety inspection.

In the 6th row, i', L, -1(8) of the fuel injection system in the current light is shown, and the m diagram is shown in Figure S. In a preferred embodiment, the device includes a housing 90 including a ventilator 14a;
14e is provided. to the air, parts 14b, 14C
around the filter 91't-, and then to the inlet 14d of the mixing chamber 20. The hydrogen passes through the quenching device 37 and enters the mixing basket 20. The nonvolatile gas enters the mixing chamber 20 through the flow path 9 and through the cooling device 139.

Note that diagram W47 shows all the mechanical components of the mixing chamber 20 that are shown separately in other diagrams.

Now, returning to Figure 1, the K11 K11 has a mixing pump 1 which is powered by the Guri 93 which is driven by the engine.
A flow path 9 for a non-volatile gas to flow through the 91 is shown, in which a valve 95 for d, Jt II k Ayu sll is arranged.
There is a pump 96 driven by r, L U
rt path 85 to the south, this flow path 85 is connected to the 11 cap filter 92 and the valve 87, and is unfortunately crowded with the main 2 UL. *Soil removal, non-oil @?
As in the case of Ilt-type engines, the oil supply is controlled by the oil supply tAr, the passage 85, and the chamber 2.
It is called within 0 and performs the beak action.

Now, in the past, there have been some research presentations on the properties of hydrogen gas, its latent uses, generation systems, and Anzensha.
tional Bureau of 5standard
s Issued 1 water = -S selected B @ = (5e
Selected Properties of Hydro
gen) (Engineering Oerlgn)
Pata).

These spinning nine light tables, as kings, generate water systems and
, which is an image line and expensive method. In addition, these research presentations are related to water-based gases that have been pirated because water-based gases are miserable and have a serious problem. suggests dangers in practical use.

^49 Doctor's graph (f, alcohol, grono41/,
It shows that the combustion exhaustion of Mej1], Sori/, Daenen Gas and Diesel oil is at least 3Sp・Ramenmanda5 [ij■. Sarari Jin, this graph shows that the combustion speed of water-based gas is from a minimum of 265 to a maximum of 325. Simply put, the speed of a commercially available fuel in g
7. ! It is about f*.

Because hydrogen gas has an unusually high combustion rate, some previous researchers have not considered hydrogen gas as a substitute for salmonids. In general, even if the engine were to be modified to VLLiM to accommodate this extra combustion rate, the idea of using it commercially due to the risk of explosion would be ruled out. Probably.

SUMMARY OF THE INVENTION The present invention, as stated above, solves the problems encountered in using hydrogen gas in standard commercially available engines. As stated in the 1st issue of I/I, the cost of water-based gas is minimal. In addition, the method described in the above-mentioned process reduces the combustion speed fJ of hydrogen gas. These U.S. applications only teach about the low points of a combustion series;
It also teaches the control of water-based power pickpocket combustion branches.

Thus, in accordance with the present invention, a practical example of applying a hydrogen generator to a combustion engine is presented as a preferred input relay. In this system, the flow of hydrogen gas is fi14m, and the non-flammable gas (oxygen) is mixed in a controlled manner and sent to the mixing chamber, thus reducing the combustion rate of water-based gas. let The device of the invention therefore makes the use of hydrogen as safe as any other.

Furthermore, as a major problem, the device of the present invention can be used even with an internal combustion engine of any size or type of reactor type, and which operates using only water as fuel. be done. Water-based gases are produced from water at very low voltages, without any chemical or chemical reactions. The combustion rate of hydrogen gas is reduced to the point where the combustion rate of the normal fuel remains. After the operation, all the parts rJ, one 'fc should have more safety valves or bulges, so that the system of the car using hydrogen gas can also be made safe. There is.

It must be understood that the terms non-volatile and non-combustible, which are used interchangeably in the above sequel, refer to the same thing and are merely intended to mean a non-flammable gas.

It is also important to note that the use of the term storage primarily in relation to the hydrogen storage area 7 is not intended to be interpreted in a literal sense; in practice, this is not storage but temporary storage. It is a retention area. This area 7 holds a sufficient amount of water for temporary starting.

Ii! Other terms, percentages, devices, etc. regarding the preferred embodiment of the invention! I made it clear. It must be understood that the present invention is susceptible to numerous modifications without departing from its technical scope.

Figure 1 shows a partial block diagram of a large preferred embodiment of the apparatus of the present invention.

#! Figure 2 shows the water jetting iron weight being removed from the example shown in @.

The third factor is the fuel mcτ11 at +1 in the example shown in Figure @1.
1! This represents a $1 type system.

FIG. 9 shows a fuel injection system for a moving wheel using the inventive device.

Figure S shows the flat surface of the fuel injection device used in this workshop.

FIG. 6 is a side sectional view of the fuel injection device of the present invention.

FIG. 7 shows II! of the fuel mixing chamber in the present invention. FIG. 1, which is a front view, is a diagram of the valve for introducing air into the fuel mixing valve according to the present invention.

No.? The figure is an integragraph of the burning rate of various slagids for explaining the present invention.

1... Water supply source, 3... Plate (excitation device), 4...
Hydrogen gas, 7...Aqueous storage area, 10°...Hydrogen generator, 20...Gas mixing chamber.

Drawing power engraving (no change in content) semi-studded / 7 empty raft middle I boke up be final law envy (c tn7s
) Procedural amendment (Form) 586.5° Showa year, month, day, case description Showa II, Permanent Application No. J647*
VSOS name Hydrogen gas injection device for internal combustion engine,
Relationship with the case of the person making the amendment Applicant's name: S179-A Meyer. Agent, mEh+oBt Showa! L capital! Month J/11, subject to correction
All simw

Patent Citations (14)

Publication number	Priority date	Publication date	Assignee	Title
JPS4946020A *	1972-09-08	1974-05-02		
Family To Family Citations				
GB591718A *	1944-11-28	1947-08-27	George Edward Heyl	Apparatus for supplementing the fuel-air mixture of an internal combustion engine by a gas mixture produced by electrolysis
US1996861A *	1929-10-28	1935-04-09	Dores Gaston	Device for preventing back-firing, capable of being used as theft-preventing devicesfor any applications
FR721354A *	1931-08-14	1932-03-02		Method and device for increasing the power of automobile engines and the like
US2109963A *	1936-08-13	1938-03-01	Alvin H Kruse	Apparatus for supplying gas to internal combustion engines
DE1021637B *	1954-03-29	1957-12-27	Botto Meccanica Di Prec E S R	Pressure carburetor for internal combustion engines with liquefied petroleum gas
US3948224A *	1970-04-22	1976-04-06	Knapp Edward M	System to prevent pollution of atmosphere
US3648668A *	1970-07-08	1972-03-14	Ebert Michael	Gas-operated internal combustion engine
JPS5241830B2 *	1975-03-05	1977-10-20		
US4031865A *	1975-10-01	1977-06-28	Patrick Dufour	Hydrogen-oxygen fuel cell for use with internal combustion engines
US3982878A *	1975-10-09	1976-09-28	Nissan Motor Co., Ltd.	Burning rate control in hydrogen fuel combustor
DE2641228A1 *	1976-09-14	1978-06-01	Rudolf Werner Schmitz	Gas mixt. powered vehicle - uses hydrogen and oxygen produced by electrolysis as fuel source
AU523583B2 *	1978-07-13	1982-08-05	Interx Research Corp.	Thiazolidine prodrugs
US4271793A *	1979-08-31	1981-06-09	Valdespino Joseph M	Internal combustion engine

* Cited by examiner, † Cited by third party

Cited By (33)

Publication number	Priority date	Publication date	Assignee	Title
Family To Family Citations				
DE3280143D1 *	1982-08-25	1990-05-03	Stanley A Meyer	DEVICE FOR GENERATING GASES FROM WATER AND FOR CONTROLLED COMBUSTION OF THESE GASES.
EP0106917A1 *	1982-10-21	1984-05-02	Stanley A. Meyer	Gas electrical hydrogen generator
CA1231872A *	1983-03-23	1988-01-26	Stanley A. Meyer	Hydrogen injection system

PL167622B1 *	1990-08-19	1995-09-30	Sheikh Bahaeddin Inst Sl	Heat generating apparatus
WO1992008046A1 *	1990-11-02	1992-05-14	Meyer Stanley A	Hydrogen gas fuel and management system for an internal combustion engine utilizing hydrogen gas fuel
US5293857A *	1990-11-02	1994-03-15	Stanley Meyer	Hydrogen gas fuel and management system for an internal combustion engine utilizing hydrogen gas fuel
JP3116169B2 *	1991-04-17	2000-12-11	本田技研工業株式会社	Fuel control method for hydrogen engine
US5178119A *	1991-12-11	1993-01-12	Southwest Research Institute	Combustion process and fuel supply system for engines
MX9206040A *	1992-03-05	1994-03-31	Southwest Res Inst	FUEL AND ADDITIVE TO REDUCE THE EMISSIONS OF PARTICLES FROM COMPRESSION IGNITION ENGINES.
US5231954A *	1992-08-05	1993-08-03	J. C. Conner	Hydrogen/oxygen fuel cell
US5501185A *	1995-01-06	1996-03-26	Kohler Co.	Biogas-driven generator set
RU2093696C1 *	1996-03-19	1997-10-20	Конструкторское бюро химавтоматики	Method for failure protection of power plants using hydrogen as working medium
JP3800020B2 *	2001-03-13	2006-07-19	日産自動車株式会社	Compression self-ignition engine
US6681749B2	2001-11-13	2004-01-27	Raymond B. Bushnell	Vapor fueled engine
US20080032245A1 *	2003-11-11	2008-02-07	Vapor Fuel Technologies, Llc	Fuel utilization
US7028675B2 *	2003-11-11	2006-04-18	Vapor Fuel Technologies, Inc.	Vapor fueled engine
US6907866B2 *	2003-11-11	2005-06-21	Vapor Fuel Technologies, Inc.	Vapor fueled engine
US20070277790A1 *	2006-06-01	2007-12-06	Raymond Bryce Bushnell	System for improving fuel utilization
US7631637B2 *	2006-06-01	2009-12-15	Vapor Fuel Technologies, Llc	System for improving fuel utilization
US8141523B2 *	2008-09-03	2012-03-27	Petrolfree, Inc.	Method and apparatus for controlling an electric motor
US8720389B2	2008-09-03	2014-05-13	Petrolfree, Inc.	Method and apparatus for controlling an electric motor and an internal combustion engine
US8336508B2 *	2008-09-17	2012-12-25	Timothy Huttner	System and method for use with a combustion engine
AU2009298157A1 *	2008-10-02	2010-04-08	David Inwald	System and method for improving combustion using an electrolysis fuel cell
US8709221B1	2010-02-24	2014-04-29	Andrew L. Smith	Current regulator for mobile water electrolyzer
US20110214593A1 *	2010-03-05	2011-09-08	Prabir Kumar Roychoudhury	Eco-friendly system and process for generating thermal energy from waste biomass
BR112016006095A2 *	2013-10-30	2017-08-01	Kumar Barthakur Jitendra	explosive exothermic implosion-type motor (ieex-em) employing a safe piping system (sps) and other safety devices
WO2016054371A1 *	2014-10-02	2016-04-07	Imagestatistics, Inc.	Voltage calculator and generator for on-board diagnostic system and method of using the same
US9771859B2	2015-10-19	2017-09-26	Christopher Haring	Hydrogen generator and control for internal-combustion vehicle
US9771658B2	2015-10-19	2017-09-26	Christopher Haring	Hydrogen generation and control for internal-combustion vehicle
CN106351741B *	2016-08-30	2018-07-31	合肥恒信汽车发动机部件制造有限公司	A kind of hydrogen comburant engine system
US20190264609A1 *	2016-11-18	2019-08-29	Hydrive Aps	Method of cleaning an internal combustion engine and system therefor
US12104560B1	2022-10-17	2024-10-01	Budhadeb Mahakul	Direct in cylinder injector for gaseous fuel
US12018631B1	2023-08-29	2024-06-25	Christopher Haring	Enhanced control of hydrogen injection for internal combustion engine system and method

* Cited by examiner, † Cited by third party, ‡ Family to family citation

Similar Documents

Publication	Publication Date	Title
JPS58202352A	1983-11-25	Hydrogen gas injector for internal combustion engine
EP0815902B1	2001-08-29	Method for lighting fire and apparatus for carrying out said process

US11092117B2	2021-08-17	System and method of improving fuel efficiency in vehicles using HHO
US6689259B1	2004-02-10	Mixed gas generator
DE112012006249B4	2016-09-01	Method for designing a working flow of a piston gas engine with spark ignition
DE10202172A1	2003-08-07	Process for disposing of boil-off gas from a cryogenic tank and motor vehicle operated in this way
EP2439395B1	2018-05-09	Device for pressurising a lift tank
DE2816115A1	1978-11-02	METHOD AND DEVICE FOR OPERATING COMBUSTION ENGINES
DE102021004278A1	2023-02-23	Blanket Pistol
EP1414663A1	2004-05-06	Method for operating a motor vehicle fuel tank system, especially a cryotank system and corresponding tank system, for example, for liquid hydrogen
CA1231872A	1988-01-26	Hydrogen injection system
EP0111574B1	1991-09-11	Combustion system for mechanical drive systems using gaseous hydrogen as fuel
Al-Khaledy	2008	High performance and low pollutant emissions from a treated diesel fuel using a magnetic field
JPS5765850A	1982-04-21	Vapor lock prevention unit
JPS59153922A	1984-09-01	Hydrogen aeration type injector
JPS5778491A	1982-05-17	Liquefied petroleum gas-containing gas oil fuel for diesel engine
DE3630345C2	1992-11-05	
DE19704587A1	1998-09-24	Emergency ballast tank blowing device for submarine
EP0187212B1	1988-06-01	Power system independent of the ambient air for burning propellant combinations
DE566436C	1933-11-01	Internal combustion engine with mixture amplification by adding hydrogen or oxyhydrogen gas to the charge
CA1227094A	1987-09-22	Hydrogen/air and non-cumbustible gas mixing combustion system
KR20240094365A	2024-06-25	The system for generating the energy
RU2083856C1	1997-07-10	Plant for producing hydrogen and oxygen from water for feeding internal-combustion engines
DE3545049A1	1987-06-25	USE OF HYDROGEN PEROXIDE AS A FUEL AND / OR OXYGEN SUPPLIER IN PISTON PISTON ENGINES
CN107701333A	2018-02-16	One kind processing water injection system

Priority And Related Applications

Applications Claiming Priority (2)

Application	Filing date	Title
US06/349,185	1982-02-17	Hydrogen gas injector system for internal combustion engine
US349185	1994-12-07	

Concepts

machine-extracted

Download Filter table

Name	Image	Sections	Count	Query match
Hydrogen		title,claims,abstract,description	79	0.000
combustion reaction		title,claims,abstract,description	62	0.000
gas		claims,abstract,description	106	0.000
mixing		claims,abstract,description	37	0.000
hydrogen		claims,abstract,description	35	0.000
hydrogen		claims,abstract,description	35	0.000
fuel		claims,abstract,description	20	0.000

oxygen	claims,abstract,description	5	0.000
oxygen	claims,abstract,description	5	0.000
water	claims,description	39	0.000
air	claims,description	23	0.000
mixture	claims,description	11	0.000
cooling	claims,description	6	0.000
injection	claims,description	6	0.000
injection	claims,description	6	0.000
ambient air	claims,description	5	0.000
Dioxygen	claims,description	3	0.000
quenching	claims,description	3	0.000
quenching effect	claims,description	3	0.000
regulatory effect	claims	2	0.000
L-glutamine	claims	1	0.000
natural gas	claims	1	0.000
natural water	claims	1	0.000
oxygen atom	claims	1	0.000
rolling process	claims	1	0.000
spray	claims	1	0.000
atomic oxygen	abstract,description	4	0.000

Show all concepts from the description section